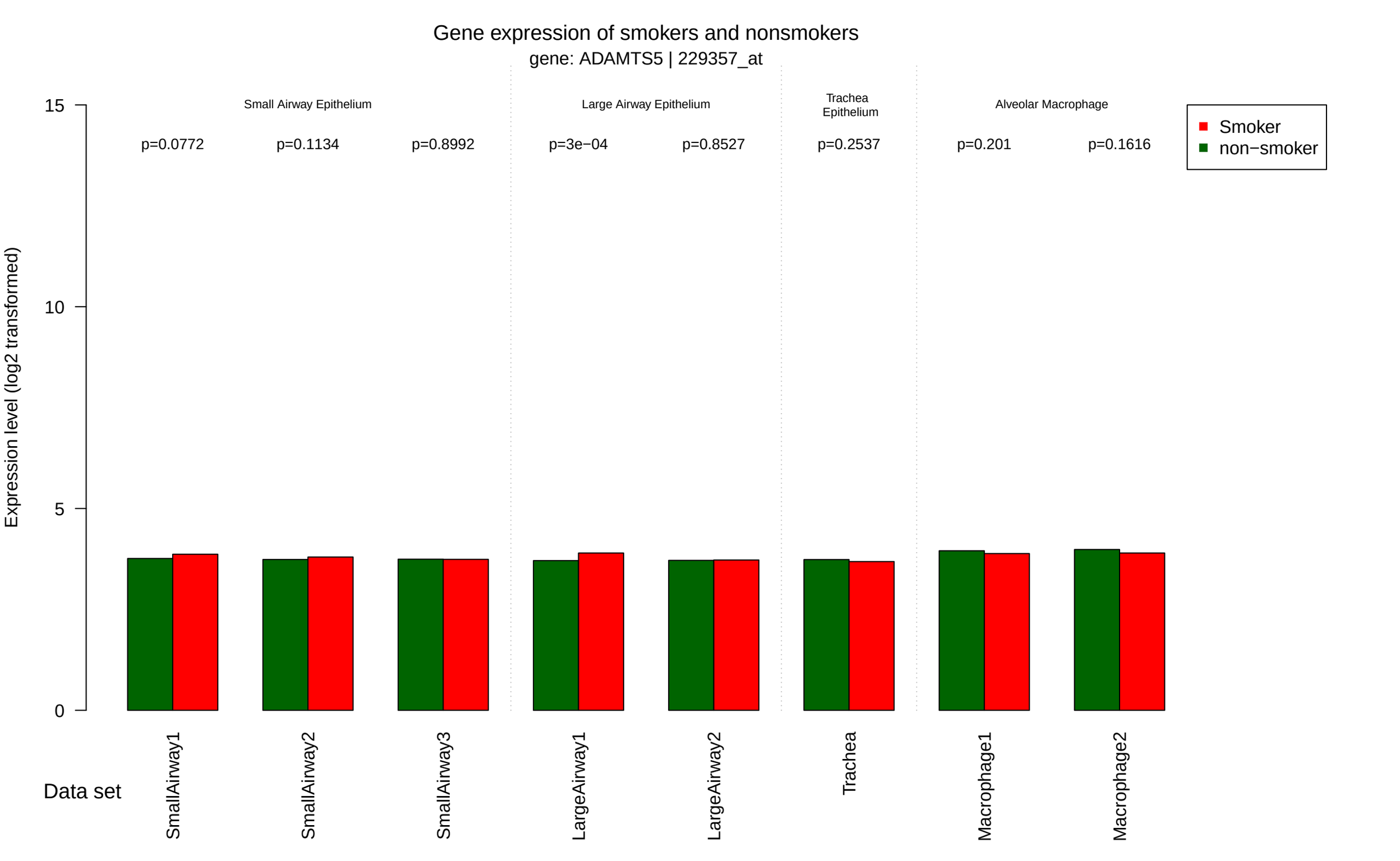
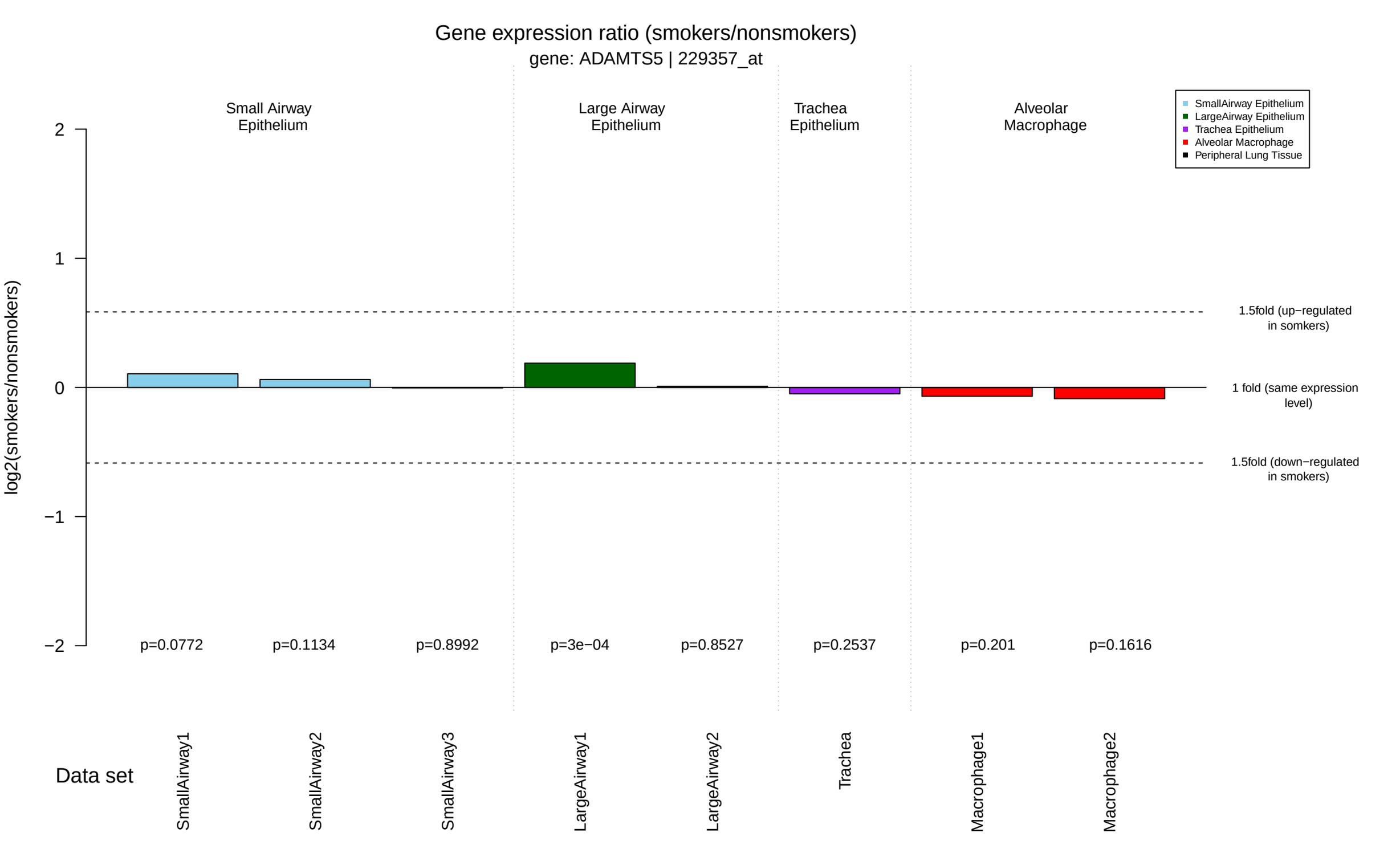
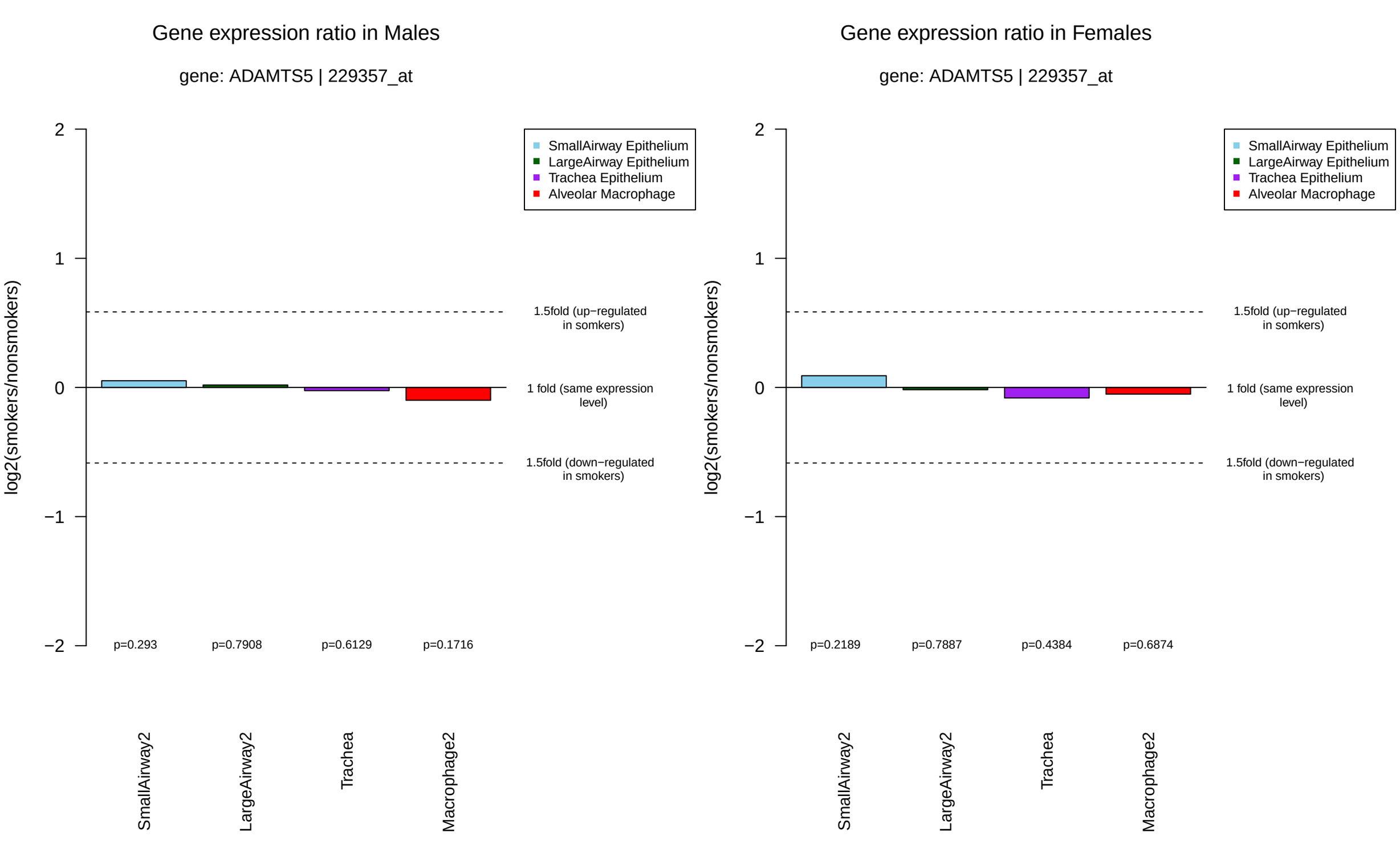
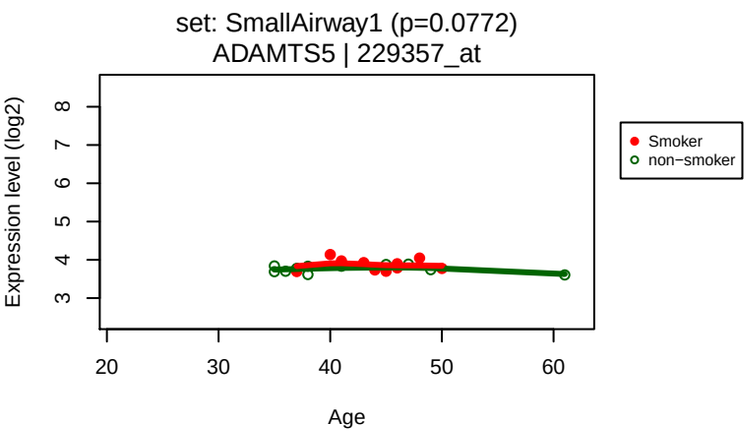
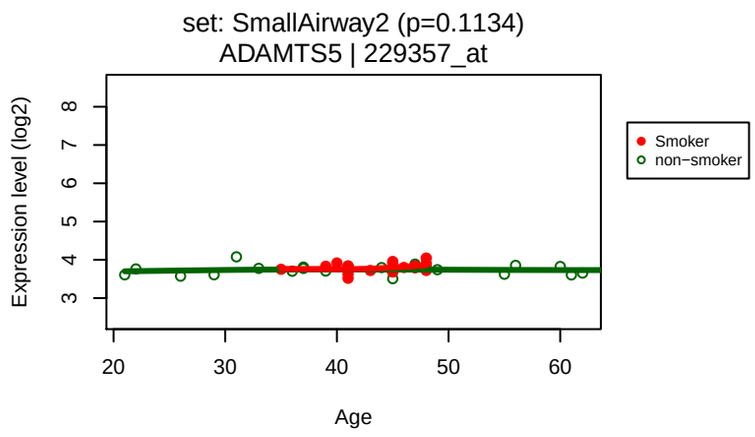
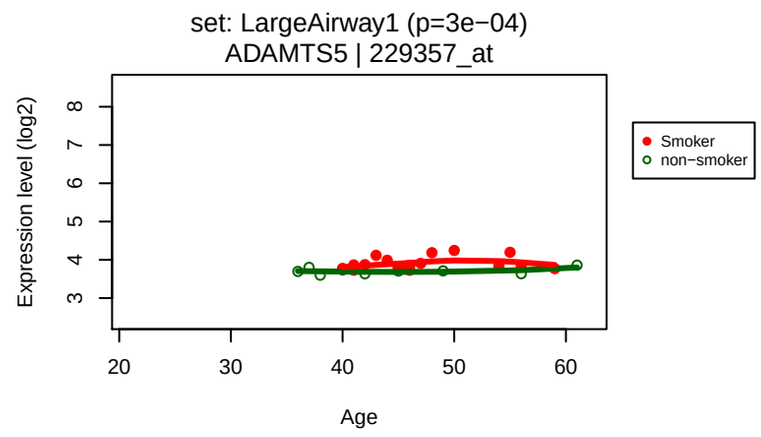

Gene expression of smokers and nonsmokers
gene: ADAMTS5 | 229357_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0772
p=0.1134
p=0.8992
p=3e−04
p=0.8527
p=0.2537
p=0.201
p=0.1616
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

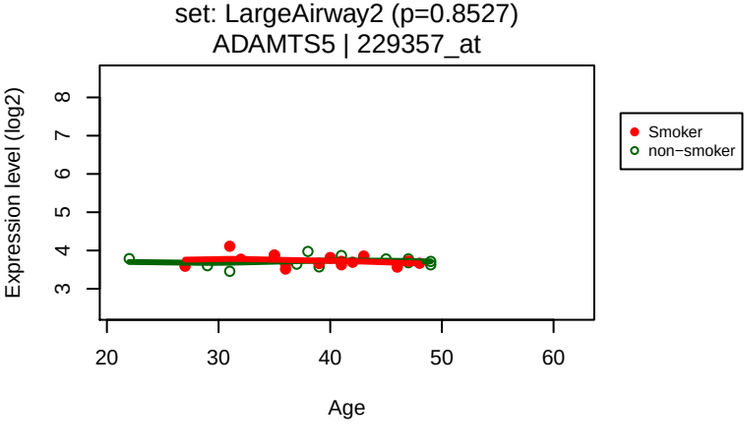

Gene expression ratio (smokers/nonsmokers)
gene: ADAMTS5 | 229357_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0772
p=0.1134
p=0.8992
p=3e−04
p=0.8527
p=0.2537
p=0.201
p=0.1616
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

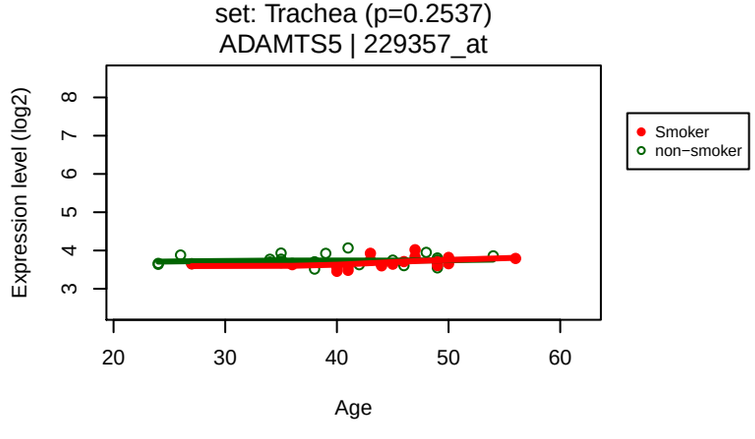

Gene expression ratio in Males
gene: ADAMTS5 | 229357_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.293
p=0.7908
p=0.6129
p=0.1716
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ADAMTS5 | 229357_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2189
p=0.7887
p=0.4384
p=0.6874
SmallAirway2
LargeAirway2
Trachea
Macrophage2

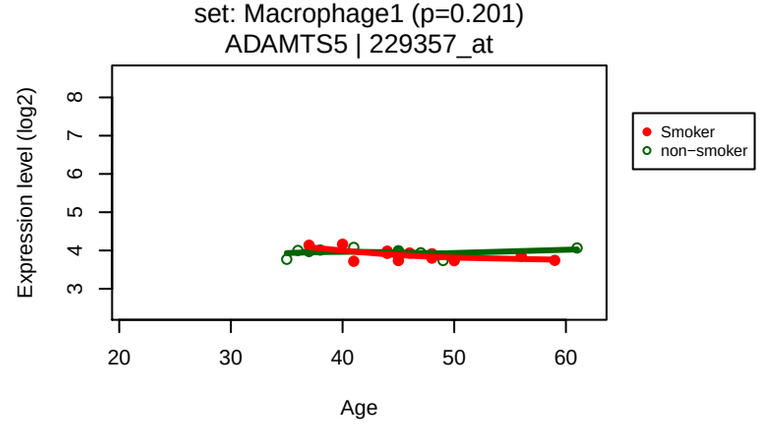

set: SmallAirway1 (p=0.0772)
ADAMTS5 | 229357_at
Expression level (log2)
Age
Smoker
non−smoker

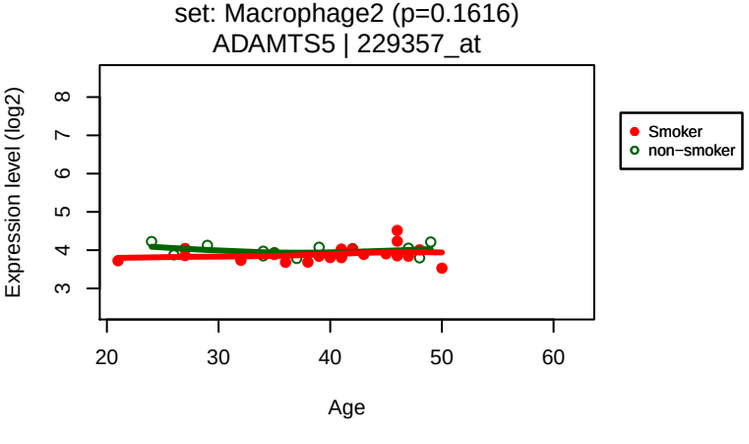

set: SmallAirway2 (p=0.1134)
ADAMTS5 | 229357_at
Expression level (log2)
Age
Smoker
non−smoker

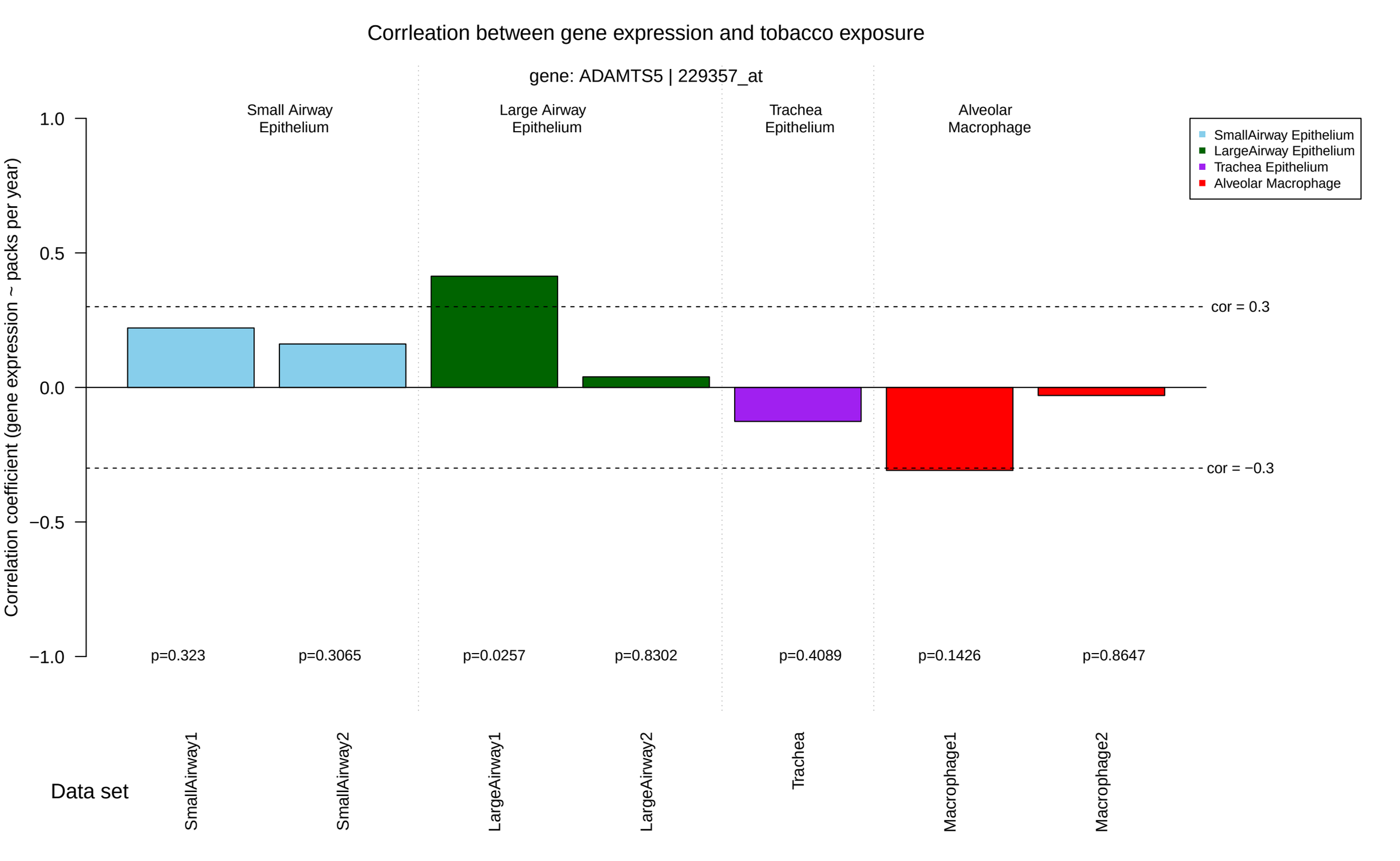

set: LargeAirway1 (p=3e−04)
ADAMTS5 | 229357_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8527)
ADAMTS5 | 229357_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2537)
ADAMTS5 | 229357_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.201)
ADAMTS5 | 229357_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1616)
ADAMTS5 | 229357_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ADAMTS5 | 229357_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.323
p=0.3065
p=0.0257
p=0.8302
p=0.4089
p=0.1426
p=0.8647
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2